Data Science Lab Project

Digital Financial Literacy: A Multidimensional Posetic Analysis of the 2023 Italian Population

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Abstract

Digital financial literacy (DFL) is essential for navigating online financial systems securely. This study investigates the structural bottlenecks preventing the Italian population from achieving secure digital financial profiles across socio-demographic groups. Using microdata from the 2023 IACOFI survey, a multidimensional framework based on poset theory is applied to the OECD/INFE dimensions of knowledge, behaviour, and attitude. The results show that attitude and knowledge represent the main structural constraints, whereas behavioural practices appear comparatively more developed. In particular, the modal profile $(0, 1, 0)$, representing 25.5% of the weighted population, reveals a structural mismatch in which intermediate behavioural practices coexist with weak cognitive and attitudinal foundations. Logistic regression indicates that financial self-confidence is negatively associated with behavioural mismatch, whereas socio-demographic background plays a limited role. At the same time, DFL profiles remain stratified across the population: low educational attainment nearly quadruples the probability of belonging to the highly vulnerable profile $(0, 0, 0)$, a persistent North-South divide emerges, and younger cohorts show stronger knowledge levels but weaker behavioural practices than older individuals.

Keywords: Digital Financial Literacy, Behavioural Mismatch, Poset Analysis, Financial Vulnerability, Multidimensional Policy Interventions

Dataset: Surveys on Financial Literacy and Digital Financial Skills in Italy: Adults - 2023 (Banca d'Italia, 2023 [4]).

AI tools: Assistance with optimizing the code implementation of the mutual ranking probabilities algorithm.

Contents

1 Research Question

1.1 Statement

Since 2017, the Bank of Italy has been conducting triennial surveys on financial literacy in Italy. Only in the 2023 edition were questions on digital financial skills introduced for the first time, creating a unique opportunity to analyse a dimension of financial proficiency that previous studies were unable to capture. In this context, digital financial literacy has emerged as a fundamental capability. It encompasses the knowledge, skills, and attitudes that enable individuals to safely and effectively use digital financial services, combining elements of traditional financial literacy and digital literacy while also accounting for the risks specific to digital financial environments (Lyons & Kass-Hanna, 2021 [11]). Following the OECD/INFE framework (OECD, 2022 [13]), digital financial literacy is decomposed into three dimensions: knowledge, behaviour, and attitude. Knowledge is assessed through questions related to the understanding of digital financial contracts, awareness of the commercial use of personal data, and comprehension of crypto-assets. Behaviour refers to the protection of passwords and PINs, verification of financial providers, management of sensitive financial information shared online, and regular password updating. Attitude reflects individuals' predisposition toward online financial security, involving practices such as avoiding public Wi-Fi networks for financial transactions, checking website security before online purchases, and reading terms and conditions.

1.2 Question

This paper aims to identify the structural bottlenecks that prevent individuals from reaching a secure digital financial profile in the digital age, a particularly relevant issue in Italy, where DFL has not yet been extensively documented. The central question guiding this study is: *Which are the primary structural bottlenecks to digital financial security in Italy, and how are they shaped by the interaction of individual profiles and behavioural drivers?* To address this overarching objective, the study develops three complementary analytical perspectives that jointly contribute to the same research framework: *Which dimension of digital financial literacy acts as the critical constraint preventing the Italian population from reaching a secure benchmark profile? What factors mediate the translation of digital behaviours into financial knowledge and secure attitude? How do regional disparities, generational gaps, and educational levels shape DFL profiles?*

1.3 Motivation

To motivate these research questions and clarify the relevance of digital financial literacy in the Italian context, it is necessary to consider the structural features of financial literacy and their interaction with the vulnerabilities of an increasingly digital financial environment.

Over the last two decades, Italy's persistent financial literacy deficit has been extensively documented, with empirical evidence revealing limited understanding of fundamental financial concepts and strong inequalities across socio-demographic and territorial groups, placing the country as an outlier among high-income economies (Di Salvatore et al., 2018 [7], Baglioni et al., 2018

[3]). Early evidence from Italian households has shown that most individuals lack basic financial concepts, such as interest rate and inflation, which negatively affects retirement planning behaviour, and that these deficiencies are concentrated among women, younger generations, less-educated individuals, and residents of Southern Italy and the Islands (Fornero & Monticone, 2011 [10]). In particular, despite being a relatively high-income country, Italy exhibits literacy levels more similar to those of medium-income economies, a pattern reflected in low participation in supplementary pension schemes, excessive reliance on dominated financial products, and limited investment experience, which restricts portfolio diversification (Bajo et al., 2015 [8]). Consequently, these vulnerabilities are not merely a competency gap but have tangible welfare implications, as recent evidence links financial illiteracy to over-indebtedness and inability to face financial and economic shocks (Bottazzi & Oggero, 2023 [5]).

Against this backdrop, the last decade has also witnessed major transformations in the financial sector. The diffusion of digital financial technologies, including online banking, mobile payments, and algorithmic advisory platforms, has reshaped the interface between households and financial services (Feyen et al., 2021 [9]). This shift introduces additional complexity on top of traditional financial concepts, redefining the competencies required to make secure financial decisions. For instance, recent evidence from the Italian population shows that individuals with higher financial literacy tend to avoid automated advisory services, whereas subjective financial overconfidence, particularly prevalent among the digitally inexperienced, drives reckless engagement with more complex tools (Aristei & Gallo, 2026 [1]). Therefore, limited digital financial literacy may amplify rather than mitigate the risks already documented in the traditional financial literacy literature. Despite these developments, existing studies on Italy typically rely on traditional financial indicators and overlook dimensions that may represent critical bottlenecks in the digital age. In fact, the digital dimension introduces additional heterogeneity, as new predictors such as financial self-confidence and familiarity with digital payment instruments coexist with classical socio-demographic factors. To account for this complexity, this study adopts a multidimensional framework aimed at analysing how digital financial vulnerabilities emerge from the interaction between socio-demographic factors, behavioural patterns, and heterogeneous DFL profiles.

1.4 Stakeholders

This research may be relevant for institutions involved in the design, regulation, and supply of digital financial services. As digital financial technologies become increasingly widespread within the financial sector, understanding how knowledge, behaviour, and attitude jointly shape digital financial security becomes essential to assess both the opportunities and risks associated with digital financial environments (OECD, 2017 [12]). First, public institutions and policymakers may benefit most directly from identifying the most significant structural bottlenecks limiting digital financial security and understanding how these vulnerabilities vary across territorial, generational, and educational settings. In particular, these findings may support the design of more targeted interventions, helping determine whether public funding should prioritize theoretical financial education, digital training, or cybersecurity awareness initiatives aimed at reducing the digital gap and exposure to digital financial risks. Moreover, national cybersecurity and regulatory bodies may also benefit from the identification of the mechanisms underlying user vulnerability in digital financial environments. Analysing the role of behavioural drivers and financial self-confidence may contribute to the development of preventive strategies against digital fraud and improve the protection of more exposed population groups. Finally, financial service providers may use the identified digital financial literacy profiles to enhance consumer protection and trust in digital financial services. Understanding heterogeneous vulnerabilities may support the design of digital environments that are safer, more accessible, and better aligned with users' behavioural and informational limitations.

2 Data

2.1 Data Source and Variables

This study is based on microdata from the Survey on Financial Literacy and Digital Financial Skills in Italy ("Indagine sull'Alfabetizzazione Finanziaria e sulle Competenze Digitali degli Italiani" - IACOFI), conducted by the Bank of Italy in 2023. The survey represents the Italian implementation of the OECD/INFE framework for measuring financial literacy and financial inclusion. In particular, the raw dataset consists of 4,862 individual observations referring to the Italian population aged between 18 and 79. To ensure population representativeness and account for the survey design, each record is associated with a sampling weight. Following the OECD/INFE guidelines, the weighting scheme is essential to adjust for selection probabilities and align the sample with the demographic structure of the Italian population in terms of gender, age distribution, and geographical area.

According to the research objectives, the analysis focuses on the dimensions of digital financial literacy identified within the OECD/INFE methodological framework. Therefore, a subset of variables is extracted from the original survey in order to construct the three core indicators of digital financial literacy, namely knowledge, behaviour, and attitude. These dimensions are derived from survey questions related to digital financial understanding, online security practices, and attitudes toward digital financial risks. The analysis also retains information on financial self-confidence, measured through respondents' self-rated financial knowledge compared to their peers, and the use of digital financial services, providing additional insight into individuals' exposure to digital environments. To further contextualize the Italian population, the dataset is complemented with a set of socio-economic characteristics, namely gender, age group, educational attainment, household income, macro-regional area, living community, and internet access. These indicators are aggregated into ordinal categories in order to construct comparable multidimensional profiles and support the analysis of heterogeneity within the Italian population.

2.2 Population Overview

The weighted distribution of the sample reflects the demographic structure of the Italian population, which is almost balanced in terms of gender, with females representing approximately 51% of the population. The age distribution is mainly concentrated among individuals aged between

55 and 79 (40%), followed by adults aged between 35 and 54 (38%), while younger individuals aged between 18 to 34 account for 22% of respondents. According to educational attainment, the majority of respondents hold a high school diploma (56%), while approximately 25% have at most a middle school qualification and around 19% possess a university degree or higher qualification. Geographically, the weighted sample is mainly concentrated in Northern Italy (46%), followed by Southern regions and islands (34%), while approximately 20% reside in Central Italy. In addition, above half of the population lives in urban areas with more than 100,000 inhabitants, while around 41% resides in medium-sized towns. With respect to household income, most respondents belong to low-income or middle-income groups, with approximately 45% reporting a monthly household income below $1,750 and 48% between $1,751 and $2,900, whereas only 7% of the population belongs to the highest income category. Lastly, internet access is widespread across the sample, involving nearly 89% of the population.

2.3 Strengths and Weaknesses of the Data

The dataset offers several advantages with respect to the objectives of the study. First, the survey is designed according to the OECD/INFE framework, ensuring consistency with internationally recognized definitions of financial literacy and digital financial skills. In particular, the availability of distinct questions related to specific dimensions of digital financial literacy, including knowledge, behaviour, and attitude, makes it possible to adopt a multidimensional approach and explore the heterogeneity of profiles within the Italian population. Another important advantage lies in the integration of socio-demographic and territorial information, allowing for the analysis of potential disparities in digital financial literacy across different population groups.

However, the dataset also presents some limitations. Certain variables exhibit high percentages of invalid or missing responses, particularly those related to household income or advanced financial topics. For example, questions concerning digital financial contracts and crypto-assets show invalid response rates slightly above 30% and 40%, respectively, potentially reflecting limited familiarity with financial matters. During preprocessing, responses corresponding to refusal, non-applicability, and uncertainty are recoded as missing values. For knowledge-related questions, these observations are considered incorrect answers, while for behavioural and attitudinal indicators measured using Likert scales, missing values are imputed using the neutral category, preserving the ordinal structure of the variables. Another limitation derives from the self-reported nature of behavioural and attitudinal information, as stated practices may not fully correspond to actual behaviours possibly due to social desirability bias or overconfidence effects.

3 Methodology

3.1 Logic Thread of the Study

The study investigates digital financial literacy as a multidimensional phenomenon shaped by the interaction between financial knowledge, online behaviour, security attitudes, and socio-demographic conditions. The analysis is organized around a common research question aimed at understanding the main drivers of digital financial literacy in Italy. To address this purpose, three complementary analytical perspectives are developed. The first investigates the structural weaknesses limiting the achievement of digital financial security. The second examines the mechanisms linking financial knowledge, digital practices, and security awareness. The third explores territorial, demographic, and social disparities associated with different digital financial literacy configurations. In this context, reducing digital financial literacy to a unidimensional score would risk masking important differences within the Italian population. Therefore, all components of the study share a common framework based on multidimensional profiles derived from the OECD/INFE dimensions of knowledge, behaviour, and attitude, preserving the complexity and heterogeneity of digital financial literacy. Overall, this strategy is driven by the idea that strengths in one area should not compensate for weaknesses in others.

3.2 Methodological Tools

3.2.1 Profile Construction and Poset Analysis

The analysis starts from the construction of multidimensional digital financial literacy profiles represented as triples (K, B, A) of the three OECD/INFE foundational dimensions: knowledge, behaviour, and attitude. These three concepts are constructed as ordinal indicators derived from the selected survey variables. In particular, knowledge is obtained from objectively assessed questions and evaluated according to the correctness of responses, while behaviour and attitude are defined by recoding ordinal survey responses to distinguish secure from non-secure digital financial practices. Moreover, to ensure consistency and interpretability across dimensions, all indicators are mapped onto a common ordinal scale ranging from 0 to 2, where higher values correspond to more secure digital financial configurations. The scaling procedure reduces the original set of configurations to 27 profiles corresponding to all possible combinations of the three dimensions. In this context, profiles are analysed through partially ordered set (poset) methodology, which compares them according to the component-wise order relation. In particular, a profile dominates another if it exhibits equal or higher levels across all dimensions.

Unlike traditional approaches that tend to derive a single aggregate score, poset analysis preserves the multidimensional structure of digital financial literacy and avoids compensating vulnerabilities in specific dimensions through strengths in others. This aspect is particularly relevant in the context of digital financial literacy, which is characterized by potentially unbalanced configurations across knowledge, behaviour, and attitude. First, the partial order structure is represented through the Hasse diagram, where each node corresponds to a DFL profile and each edge identifies a direct dominance relation between two comparable profiles. Then, mutual ranking probabilities are computed in order to estimate the relative positioning of profiles within the partial order structure. In particular, for each pair of profiles, the procedure evaluates the probability that one profile precedes the other across a set of linear extensions compatible with the poset (De Loof, 2009 [6]). Since the exact enumeration of all compatible rankings rapidly becomes computationally infeasible, the estimation relies on an approximation algorithm inspired by the framework implemented in the `poseticDataAnalysis` package (Avellone et al., 2026 [2]). In this case, the computation is based on a Markov Chain Monte Carlo (MCMC) approach following the Bubley-Dyer procedure for the sampling of linear extensions. To ensure convergence and obtain reliable estimates of the mutual ranking probabilities, the number of iterations is determined according to the theoretical upper bound:

$$n_\epsilon = |E|^4(\ln(|E|))^2 + |E|^3 \ln(|E|)\ln(\epsilon^{-1}),$$

where $|E|$ denotes the number of profiles within the poset and ϵ controls the maximum approximation error of the simulated distribution. Ultimately, additional indicators are derived in order to further interpret the profile space. More specifically, dominance quantifies the extent to which each profile structurally prevails over the other configurations within the partial order, incomparability measures the degree to which profiles cannot be unambiguously ordered with respect to the others, and separation evaluates the distance between each pair of profiles through a fuzzy dominance framework, making it possible to identify the dimension acting as the main bottleneck with respect to the benchmark configuration.

3.2.2 Behavioural Mismatch and Logistic Regression

The poset analysis establishes the structural position of each DFL profile within the partial order, but it does not explain which individual characteristics are associated with specific configurations of vulnerability. Therefore, the study also focuses on a particular asymmetric profile that naturally emerges from the multidimensional framework: individuals whose behavioural dimension exceeds both their knowledge and attitude scores. To isolate this configuration, a binary indicator of behavioural mismatch is constructed. In particular, an individual exhibits a positive mismatch when the behaviour is strictly higher than both the knowledge and attitude dimensions. This variable captures a targeted form of structural vulnerability, in which individuals actively engage in digital financial practices without the corresponding financial understanding or attitudinal awareness necessary to operate safely within digital financial environments. To evaluate the behavioural mechanisms underlying this mismatch, additional explanatory variables are derived from the dataset. First, a digital financial services (DFS) intensity index is computed by averaging respondents' frequency of use across a range of online financial activities, including checking account balances, paying bills, transferring money, using mobile payments, and consulting automated advisory platforms. This index measures individuals' engagement with digital financial instruments and captures their degree of exposure to digital financial environments. Second, an indicator of financial self-confidence, derived from the survey item asking respondents to evaluate their overall financial knowledge compared to others, is included to assess whether more confident individuals are less prone to behavioural mismatch. Incorporating this psychological dimension makes it possible to test how self-assessed competence shapes digital vulnerability. Given the binary nature of the dependent variable, which identifies the presence or absence of the behavioural mismatch, ordinary least squares estimation would yield heteroscedastic errors and predicted probabilities outside the unit interval. To overcome these limitations and properly model the bounded probability distribution, the probability of exhibiting a positive mismatch is estimated through a generalized linear model (GLM) with binomial family. Survey weights are incorporated into the estimation process to ensure that coefficient estimates are representative of the Italian population. The baseline specification includes DFS intensity, self-confidence, gender, and internet access as regressors, alongside categorical controls for age group, education level, and household income. To avoid imposing linear effects across demographic dimensions and to capture potentially heterogenous risk configuration, these categorical variables are encoded through dummy indicators, with the lowest ordinal category serving as the baseline reference group. A second specification restricts the estimation sample to the almost 88.94% of respondents who report having internet access, excluding the internet access variable and its interactions from the model. This restriction is empirically motivated by the presence of a non-negligible share of digitally excluded individuals (11.06% of the weighted sample), for whom the behavioural indicators underlying the mismatch variable may primarily reflect limited access opportunities rather than deliberate behavioural choices. Therefore, comparing the baseline model with this restricted specification makes it possible to assess whether the identified drivers of behavioural mismatch operate uniformly across the entire population or are specific to individuals who are already actively embedded within the digital financial ecosystem.

3.2.3 Visual Analytics and Inferential Framework

In addition to the multidimensional DFL profiles, the analysis considers a set of socio-demographic characteristics aimed at capturing the heterogeneity within the Italian population. In particular, macro-region, age group, and educational level are used as the main explanatory factors associated with differences in digital financial literacy outcomes. From a theoretical perspective, age reflects life-cycle dynamics and familiarity with digital tools, educational attainment acts as a proxy for human capital and cognitive resources, whereas macro-regional differences accounts for localized disparities in economic development and access to financial infrastructure. As previously outlined, these variables are aggregated into ordinal categories in order to ensure comparability across population groups. This classification also allows the analysis to account for the relative composition of socio-demographic groups within the weighted sample.

To visualize heterogeneity in digital financial literacy profiles across socio-demographic groups, separate heatmaps are constructed for each of the considered dimensions (age group, educational level, and macro-region). Each heatmap reports the distribution of all DFL profiles within each subgroup, with values expressed as row percentages summing to 100 within each category. This normalization allows for meaningful comparisons across groups with different weighted proportions. Given the presence of 27 distinct profile configurations, this representation is particularly suitable for summarising complex information while preserving readability and interpretability. To complement this descriptive evidence, additional heatmaps based on Pearson residuals are employed for each socio-demographic indicator. These residuals measure the standardized deviation between observed and expected frequencies of digital financial literacy profiles under the assumption of independence between profile distribution and group membership. They are particularly useful in this context because they allow deviations to be interpreted relative to their expected variability, making comparisons across cells independent of marginal distributions and sample size differences. In this way, positive residuals indicate over-representation, while negative residuals express under-representation, of specific profiles within each subgroup, enabling a clear identification of structural deviations from the independence benchmark. Finally, complementary descriptive summaries are used to support interpretation of the heatmap evidence, including ranked comparisons of the most frequent profiles within each subgroup and comparisons of group-level averages against the national mean for each DFL dimension (knowledge, behaviour, and attitude). These analyses evaluate whether specific socio-demographic groups exhibit values above or below the population reference level.

From a statistical perspective, alongside the analysis of Pearson residuals, a Chi-square test of independence is employed to assess whether the distribution of DFL profiles is significantly associated with socio-demographic categories. The Chi-square test is considered particularly appropriate in this context as it evaluates whether the observed distribution of profiles deviates from the distribution expected under the null hypothesis of independence between socio-demographic characteristics and DFL profile membership. In particular, to quantify the magnitude of this association, Cramér's V is calculated. This measure provides a standardized effect size for contingency tables, allowing the strength of the association between socio-demographic indicators and DFL profile distributions to be assessed independently of sample size and table dimensions. In this way, it complements the Chi-square test by distinguishing statistically significant associations from those that are also meaningful in magnitude. Lastly, a multivariate logistic regression model is also employed to further investigate the factors associated with the probability of belonging to a specific digital financial literacy profile. This approach allows the simultaneous control of multiple socio-demographic characteristics, providing a more robust assessment of their joint relationship with profile membership. In particular, the dependent variable is defined as a binary indicator identifying whether an individual belongs to the selected target profile derived from the DFL triple (K, B, A), while macro-region, age group, and education level are included as explanatory variables. A weighted generalized linear model with binomial family is estimated to account for survey weights and ensure that inference is representative of the overall population. Results are reported in terms of odds ratios with 95% confidence intervals, while statistical significance is assessed through p-values. Then, the estimated effects are used to identify the direction and magnitude of the association between socio-demographic characteristics and the likelihood of belonging to selected digital financial literacy profiles.

4 Results

4.1 Analytical Results

4.1.1 Profile Distribution, Poset Analysis, and Structural Bottlenecks

The construction of multidimensional profiles produces the complete set of 27 configurations defined by the three OECD/INFE dimensions of knowledge, behaviour, and attitude. All possible profiles are observed within the Italian population, suggesting a highly heterogeneous structure of digital financial literacy. However, considering the general descriptive statistics, there are significant differences across the three dimensions. Since each indicator is measured on a common ordinal scale ranging from 0 to 2, values can be interpreted as representing low, intermediate, and high levels of digital financial literacy, respectively. In particular, behaviour exhibits the highest average score (0.86), followed by knowledge (0.56) and attitude (0.51). The median values for both knowledge and attitude are equal to zero, suggesting that a large portion of the population is concentrated at the lowest level of these dimensions. Instead, behavioural practices appear stronger, with most individuals reaching at least an intermediate level of secure online behaviour. This asymmetry is also reflected in the most frequent profile, $(0, 1, 0)$, representing individuals characterized by low levels of digital financial knowledge and attitude, combined with intermediate behavioural practices. This configuration accounts for approximately 25.5% of the weighted population, indicating that relatively secure behavioural practices are often not supported by adequate levels of digital financial knowledge and attitude. In contrast, some profiles are extremely rare within the Italian population. For example, the configuration $(2, 0, 2)$ appears only once in the sample, corresponding to approximately 0.02% of the weighted population. This suggests that high levels of knowledge and attitude rarely coexist with weak behavioural practices.

The relationships among the 27 profiles are analysed through the poset framework, which preserves the multidimensional structure of digital financial literacy. In particular, Figure 1 visualizes the Hasse diagram of the partial order derived from the three DFL dimensions. It highlights both the benchmark profile $(1, 1, 1)$ and the modal profile $(0, 1, 0)$, allowing a direct comparison between the secure configuration and the most frequent profile in the Italian population.

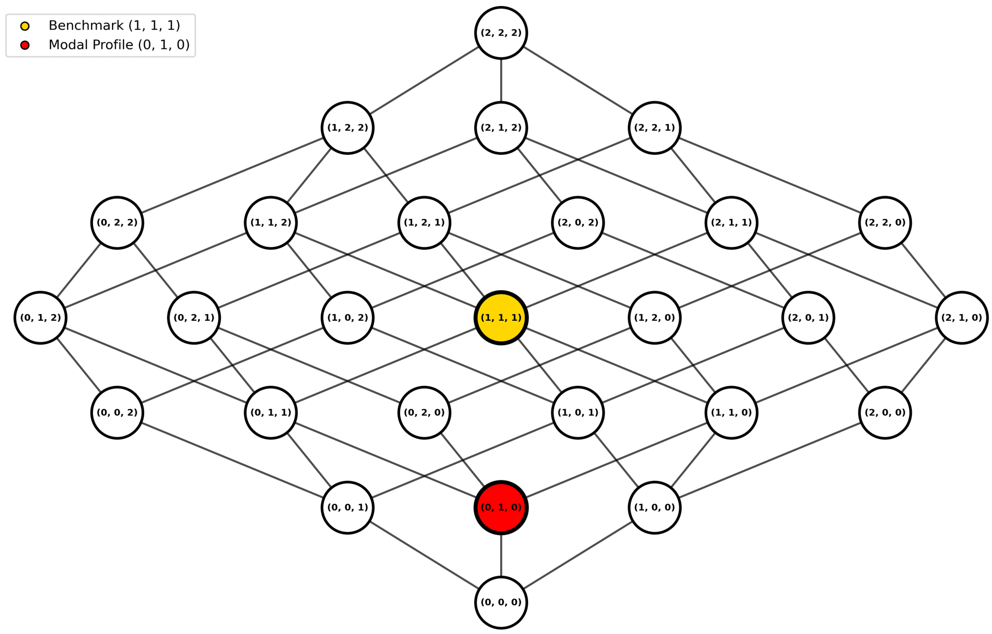

Figure 1: Hasse Diagram of Digital Financial Literacy Profiles

Starting from the partial order structure, mutual ranking probabilities are computed in order to estimate the relative positioning of profiles across a sampled set of linear extensions compatible with the poset. This probabilistic approach allows the definition of additional indicators to further describe the structure of the profile space. Within this context, the profile $(1, 1, 1)$ is identified as the benchmark configuration, representing a balanced intermediate level across all dimensions. Nevertheless, this profile involves only approximately 7.2% of the weighted population, remaining less frequent than structurally unbalanced configurations. Moreover, the benchmark profile exhibits a relatively strong dominance score (12.95) together with a moderate incomparability level (0.25), indicating its relevant role within the profile space. These values are substantially higher than those of the modal profile (2.30 and 0.10, respectively), suggesting that the benchmark occupies a central position within the partial order, whereas the modal profile corresponds to a weaker configuration. Overall, the weighted incomparability index is 0.15, indicating the presence of a portion of configurations that remain structurally ambiguous, as individuals often display heterogeneous combinations of strengths and weaknesses across the three dimensions, making a complete ranking of the population conceptually inadequate.

Then, to identify the dimension acting as the main structural bottleneck, the weighted separation from the benchmark profile is analysed for each of the three dimensions of the digital financial literacy. This indicator measures how far the population remains, on average, from the benchmark configuration in each dimension. In particular, attitude represents the largest source of vulnerability, with an average gap of 0.63 on the common ordinal scale, closely followed by knowledge (0.61), while behaviour shows a smaller deviation from the benchmark profile (0.23). Thus, individuals often display adequate behavioural practices without possessing the corresponding level of financial understanding and cybersecurity awareness required to achieve a fully secure digital financial profile. This pattern suggests that the relationship between the three dimensions may be shaped by additional underlying factors beyond simple compositional differences.

4.1.2 Regression Results of Behavioural Mismatch

The empirical estimates from the logistic regression reveal that the probability of experiencing a positive mismatch is associated with a combination of a structural access barrier and an individual psychological characteristic, rather than socio-demographic factors. Across both specifications, the pseudo R-squared values are close to zero (0.011 in the baseline and 0.003 in the restricted model), suggesting that variation in behavioural mismatch reflects a widespread baseline pattern rather than being strongly driven by traditional demographic background.

In the baseline specification, internet access emerges as a highly significant predictor ($\hat{\beta} = -0.662$, $p < 0.001$). This negative coefficient indicates that individuals with internet access have a lower probability of being classified within the strict positive mismatch profile compared to those who are completely offline. This finding is consistent with the structural motivation behind the second specification. For the offline sample of the population, the lack of internet access prevents digital behavioural engagement, thereby artificially increasing their mismatch classification due to structural exclusion rather than active behavioural choices. Furthermore, financial self-confidence emerges as a significant negative predictor ($\hat{\beta} = -0.112$, $p = 0.004$), while the digital financial services intensity index is only marginally significant ($\hat{\beta} = 0.102$, $p = 0.064$). Crucially, once the access barrier is removed in the restricted specification, focusing exclusively on the population actively embedded in the digital environment, the results highlight a specific source of heterogeneity. All socio-demographic controls such as gender, age class, education level, and household income exhibit negligible magnitudes and fail to achieve statistical significance. Furthermore, the DFS intensity index loses its marginal significance ($\hat{\beta} = 0.086$, $p = 0.125$). However, financial self-confidence remains a highly significant negative predictor ($\hat{\beta} = -0.136$, $p = 0.001$). Therefore, the restricted specification suggests that while socio-economic background does not clearly segment the risk across the connected population, individual psychological factors play a pivotal role. Rather than acting as a driver of vulnerability through unjustified self-assurance, higher financial self-confidence operates as a robust protective factor against behavioural mismatch.

4.1.3 Structural Analysis of DFL Profiles across Socio-Demographic Dimensions

Given that traditional socio-demographic variables fail to segment the risk of behavioural mismatch, the analysis is extended to a broader structural perspective. Specifically, attention is shifted from the mismatch indicator to the full distribution of DFL profiles in order to investigate whether systematic patterns emerge across socio-demographic groups.

Starting with macro-regions, the configuration $(0, 1, 0)$ is identified as the modal profile, accounting for approximately 25% in all subgroups. This indicates that an unbalanced digital financial literacy structure, characterized by intermediate behavioural practices and low levels of knowledge and attitude, represents a systemic national feature rather than a localized anomaly. However, marked regional heterogeneity is observed when focusing on other weighted profiles. In particular, the condition of absolute vulnerability, represented by the profile $(0, 0, 0)$, is found to affect 12.5% of the population in the North, increasing to 15.1% in the Center and reaching 18.0% in the South-Islands. The comparison with national averages confirms a clear geographical divide: the North is systematically at or above the national average across dimensions (with a peak in behaviour equal to 0.86), whereas the South-Islands display persistent structural weakness, remaining below the average across all dimensions. By contrast, the Center appears more balanced and broadly aligned with the national pattern. To formalize these deviations beyond descriptive frequencies, an analysis of Pearson residuals is conducted. The heatmap confirms a significant over-representation of the vulnerable profile $(0, 0, 0)$ in the South-Islands (residual = 3.3), together with a corresponding under-representation in the North (residual = -2.9). Conversely, higher-level configurations tend to display negative residuals in the South-Islands and positive residuals in the North, indicating a geographical concentration of more advanced DFL profiles. The Center exhibits a mixed pattern, with smaller and less systematic deviations from the independence benchmark. Finally, the statistical significance of these geographical differences is formally tested using the Chi-square test of independence, confirming a significant association between macro-region and the distribution of DFL profiles ($\chi^2 = 87.66$, $p = 0.001$). Cramér's V further indicates a limited association between the two variables (0.095).

Moving to the analysis of age groups, the configuration $(0, 1, 0)$ remains the most frequent across all generations, whereas the profile $(0, 0, 0)$ is most prevalent among young individuals (17.7%), compared to seniors (15.0%) and adults (13.1%). A clear behaviour-knowledge trade-off emerges across cohorts: younger individuals exhibit higher average knowledge scores (0.58) but lower behavioural practices (0.78), while older cohorts show stronger average behavioural scores (0.86 for adults and 0.83 for seniors) despite lower average knowledge scores (0.57 and 0.44, respectively). Attitude emerges as a structural weakness, particularly among young individuals (0.39), potentially reflecting lower levels of risk awareness in digital environments. The Pearson residual heatmap further confirms these patterns. In particular, young individuals are significantly over-represented in vulnerable and cognitively unbalanced profiles, such as $(0,0,0)$ (residual = 2.4) and $(2,1,0)$ (residual = 4.5). Adults are more frequently associated with balanced intermediate-to-high configurations, including $(1,1,2)$ (residual = 3.6), while seniors tend to concentrate in profiles characterized by stronger behavioural components but weaker knowledge levels, such as $(0,1,2)$ (residual = 3.2). Finally, the Chi-square test of independence confirms a significant association between age group and the distribution of DFL profiles ($\chi^2 = 177.09$, $p < 0.001$). Cramér's V also indicates a stronger association compared to the geographical indicator (0.135).

Shifting the focus to educational level, the same modal configuration $(0,1,0)$ persists across all groups. However, strong stratification emerges in vulnerability levels: the profile $(0,0,0)$ is substantially more prevalent among individuals with low education (23.6%), compared to medium (13.0%) and high education groups (8.4%). The comparison with national averages further highlights clear differences across educational levels. Indeed, highly educated individuals systematically perform above the national average across all dimensions, particularly in behaviour (0.94), while low education groups remain consistently below average, especially in knowledge (0.35) and attitude (0.36). The medium education group remains closer to the national reference values across the three dimensions. Figure 2 reports the Pearson residual heatmap for educational attainment and confirms these structural differences. In particular, the low education group exhibits a strong over-representation of highly vulnerable configurations, peaking at profile $(0,0,0)$ (residual = 13.6). In contrast, the high education group is not characterized by strong positive residuals in advanced profiles, but rather by a systematic under-representation of vulnerable profiles, including the modal configuration $(0,1,0)$ (residual = -19.0) and the vulnerable profile $(0,0,0)$ (residual = -16.4). As profile complexity increases, these negative residuals progressively weaken, indicating a closer alignment with more advanced DFL configurations. Finally, the Chi-square test of independence confirms a significant association between educational level and the distribution of DFL profiles ($\chi^2 = 294.10$, $p < 0.001$). Cramér's V indicates the strongest association among the socio-demographic indicators considered (0.174).

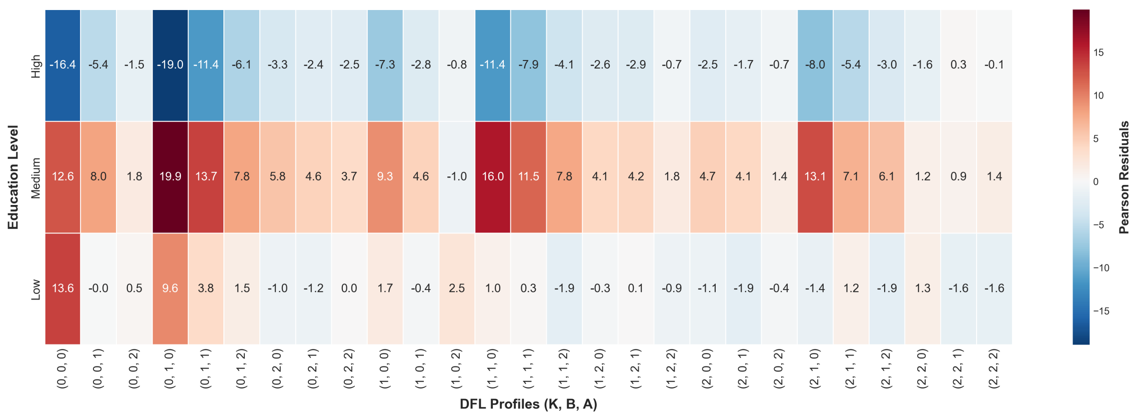

Figure 2: Educational Deviations from Expected Profile Distribution

To complement the evidence emerging from the preliminary analysis, a multivariate logistic regression model is used to estimate the joint association between socio-demographic characteristics and specific DFL configurations. In particular, the analysis focuses on the highly vulnerable profile $(0, 0, 0)$, selected because of its marked heterogeneity across socio-demographic groups, and the modal profile $(0, 1, 0)$, which represents the most frequent configuration across all categories. The results confirm that educational attainment represents the most relevant determinant. Regarding the profile $(0, 0, 0)$, individuals with a low educational level exhibit odds of belonging to this configuration that are almost four times higher than those observed among highly educated individuals (OR = 3.98, 95% CI: [2.92-5.44]). Significant geographical effects also emerge, with a higher risk detected for individuals residing in the South-Islands (OR = 1.48) and in the Center (OR = 1.31) compared to the North. After controlling for education and geographical area, an inverse age effect is identified: adults (OR = 0.78) and seniors (OR = 0.63) are less likely to belong to the fully vulnerable configuration compared to young individuals, suggesting that the latter represents the most critical group in terms of baseline fragility. The analysis of the modal profile $(0, 1, 0)$, characterized by intermediate behavioural practices but low levels of knowledge and attitude, further reinforces the central role of education. Both low (OR = 1.64) and medium educational attainment (OR = 1.20) are associated with a higher probability of remaining in this unbalanced configuration. By contrast, no statistically significant effects are detected for age group and geographical area, indicating that this profile represents a widespread and cross-cutting condition within the Italian population, primarily driven by educational background rather than generational or territorial differences.

4.2 Interpretation of the Results

The results confirm the importance of analysing digital financial literacy as a multidimensional phenomenon rather than reducing it to a composite indicator. Indeed, the observed profile distribution shows that individuals frequently exhibit heterogeneous and unbalanced configurations across knowledge, behaviour, and attitude, with strength in one dimension often coexisting with weaknesses in others. In particular, the poset framework suggests that behavioural practices tend to be relatively more developed within the Italian population, whereas attitude and knowledge remain the main constraints preventing individuals from achieving more secure digital financial profiles. This pattern indicates that the adoption of relatively secure digital financial behaviours does not necessarily imply an adequate understanding of financial risks and cybersecurity principles. From a broader perspective, these findings support the idea that improving digital financial literacy requires multidimensional policy interventions that extend beyond simple behavioural guidance and place greater emphasis on financial knowledge, critical awareness, and cybersecurity understanding. At the same time, the presence of structurally heterogeneous profile configurations suggests that the determining factors of digital financial literacy may differ across territorial, demographic, and social groups, motivating further investigation into the mechanisms underlying these disparities within the Italian population.

Consequently, the regression results regarding behavioural mismatch represent an important empirical finding that directly complements the poset analysis. While the separation indicators identify attitude and knowledge as the primary bottlenecks within the Italian digital literacy landscape, the regression analysis suggests that this asymmetric vulnerability cannot be clearly localized within specific demographic or behavioural groups. Therefore, the mismatch appears to represent a widespread phenomenon that is not systematically associated with traditional socio-demographic characteristics. Nevertheless, the risk is strongly segmented by individual psychological attributes. More specifically, the significant negative coefficient associated with internet access further highlights the presence of a structural divide. Individuals who do not access the internet are more likely to be classified within the positive mismatch category, suggesting that their behavioural responses may reflect contextual constraints or institutional guidance rather than genuine attitudinal and cognitive foundations. Once this structural divide between users and non-users is accounted for, no additional socio-demographic differentiation emerges, confirming that the mismatch phenomenon operates uniformly among digitally connected individuals. Crucially, the significant negative coefficient associated with financial self-confidence runs counter to the expectation that unjustified self-assurance universally drives vulnerability. While recent literature utilizing the same IACOFI 2023 dataset demonstrates that subjective financial overconfidence acts as a driver for the adoption of specific automated services like robo-advisors (Aristei & Gallo, 2026 [1]), our findings suggest that general financial self-confidence may operate differently at the structural level. Rather than encouraging reckless engagement, financial confidence appears to act as a mitigating factor against behavioural mismatch, potentially reflecting a greater ability to align digital financial behaviours with the underlying knowledge and attitudes required to navigate digital financial environments securely.

This result can be further interpreted in light of the descriptive evidence observed within the contemporary Italian digital ecosystem, which helps explain why individual-level predictors fail to segment the risk. A prominent 62.7% of the sample does not correctly understand digital contract terms and conditions, while 49.1% report limited awareness regarding the use of their personal data online. However, despite this cognitive deficit, individuals continue performing online financial behaviours: 33.8% execute online purchases without verifying website security protocols, and 23.4% rarely or never adopt basic password protection practices for financial accounts. This dynamic further amplifies in complex financial environments, where 66.3% of respondents lack an adequate understanding of crypto-assets and related risks, while an alarming 10.0% actively share their banking passwords or PINs. Considered together with the regression results, these percentages help explain why socio-demographic coefficients converge toward zero and the pseudo R-squared values remain extremely limited. These vulnerabilities do not appear as isolated anomalies affecting specific social groups, but rather as widespread characteristics of the Italian digital landscape. In other words, the digital dimension appears to distribute this specific form of vulnerability uniformly across social groups, pointing toward architectural and environment-level explanations rather than individual deficiencies. Consequently, policy interventions targeting digital behavioural mismatch may be less effective if based exclusively on traditional socio-demographic profiling strategies commonly adopted in conventional financial literacy programs. Instead, regulatory and educational frameworks should address the structural architecture of digital financial environments and interfaces that facilitate transactions without necessarily requiring pure user awareness or understanding.

Although the risk of behavioural mismatch appears to operate relatively uniformly across the population, it is primarily associated with individual psychological factors rather than with demographic background. By contrast, the overall composition of DFL profiles remains stratified across socio-demographic groups. Indeed, by shifting the focus from the mismatch phenomenon to the broader distribution of digital financial literacy, significant structural differences emerge across geography, age, and education through both bivariate and multivariate analyses. First, a statistically significant association between macro-region and DFL profiles is confirmed, although the magnitude is relatively modest compared to the other socio-demographic factors considered. The higher incidence of the most vulnerable profile $(0, 0, 0)$ in the South-Islands, contrasted with the greater resilience observed in the North, suggests that geographical disparities may reflect broader differences in local economic development and digital infrastructure. Moreover, a critical generational behaviour-knowledge paradox also emerges from the analysis. Contrary to the common assumption that digital natives are inherently more digitally and financially literate, younger individuals display a higher exposure to total vulnerability. Although superior cognitive foundations are observed among young and adult cohorts, a systematic lag in practical implementation and long-term financial orientation remains evident. Conversely, the theoretical disadvantages observed among seniors coexist with more consolidated and risk-averse financial habits developed through life experience. These findings highlight that digital financial literacy is not determined solely by digital familiarity, but also by the ability to translate theoretical knowledge into secure and informed financial behaviour. However, the regression results indicate that geography and age alone do not fully account for these inequalities, as educational attainment emerges as the socio-demographic factor most strongly associated with digital financial literacy profiles. In particular, higher educational level appears less associated with the concentration in highly advanced configurations than with a systematic reduction in the probability of belonging to the most vulnerable profiles. In other words, education primarily acts as a protective factor against severe digital financial vulnerability rather than as a guarantee of fully developed digital financial literacy.

5 Conclusion

5.1 Gap between the Goal and the Results

The primary objective of this research is to identify the structural bottlenecks limiting digital financial security in Italy and analyse how these vulnerabilities are shaped by the interaction between multidimensional digital financial literacy profiles and socio-demographic characteristics. Overall, this goal is largely achieved within the limits imposed by the adopted methodological framework. By applying a multidimensional poset approach to the IACOFI 2023 survey, the research preserves the heterogeneity of digital financial literacy configurations and avoids reducing individuals to a single aggregate indicator. In particular, attitude and knowledge are identified as the main systemic constraints within the Italian population, whereas behaviour appears comparatively more developed. These findings show that relatively secure digital financial practices are not necessarily accompanied by adequate theoretical understanding and cybersecurity awareness.

The regression analysis on behavioural mismatch further reveals that this asymmetry operates as a cross-cutting phenomenon, while still being associated with individual psychological traits. Once internet access is taken into account, traditional socio-demographic variables are generally not found to significantly predict behavioural mismatch, suggesting that the tendency to adopt apparently secure behaviours without a corresponding level of understanding represents a generalized feature of the connected population. However, the analysis identifies financial self-confidence as a significant individual-level factor. This suggest that, while behavioural mismatch cannot be meaningfully explained through traditional demographic characteristics, it is associated with differences in individual confidence and psychological readiness to navigate digital environments.

By contrast, when the broader DFL profile space is examined, socio-demographic characteristics are found to play a more relevant role in shaping baseline vulnerability configurations. In particular, educational attainment emerges as the factor most strongly associated with digital financial vulnerability, followed by age-related differences and geographical disparities. At the same time, the results suggest that higher levels of education are less associated with the concentration in highly advanced profiles than with a lower probability of belonging to the most vulnerable configurations. While these findings help identify the main structural determinants of digital financial vulnerability, they also indicate that part of the heterogeneity observed across profiles may depend on additional unobserved contextual factors not directly captured within the adopted framework.

5.2 Strengths and Weaknesses of the Study

This study presents several methodological and empirical strengths. A core point lies in the multidimensional analytical framework, through which the complexity of digital financial literacy is preserved by explicitly modelling knowledge, behaviour, and attitude as distinct but interrelated dimensions. In this way, the information loss typically associated with aggregated scoring approaches is avoided, allowing a more refined representation of individual heterogeneity. Moreover, the integration of poset methodology and mutual ranking probabilities provides a rigorous framework for the analysis of partially ordered structures. This approach also makes it possible to identify structural dominance relations and incomparability across profiles, generating insights that extend beyond conventional descriptive statistics and composite indices. Then, the integration of survey weights together with OECD/INFE-aligned definitions ensures methodological consistency, improving the representativeness and policy relevance of the findings for the Italian adult population. Finally, the combined use of descriptive, inferential, and multivariate techniques strengthens the robustness of the empirical analysis by capturing both profile-specific vulnerabilities and broader structural patterns.

Despite these strengths, some limitations are also identified. First, the cross-sectional nature of the 2023 IACOFI dataset prevents the analysis of temporal dynamics and limits the possibility of observing how digital financial literacy profiles evolve over time. In addition, the multidimensional structure of the framework requires a partial reduction of complexity through the aggregation of individual indicators into higher-level dimensions prior to profile construction, in order to preserve interpretability and statistical stability. While methodologically necessary, this process may limit the investigation of more detailed territorial differences or generational heterogeneity. Ultimately, although the behavioural mismatch analysis provides an important conceptual insight, the limited explanatory power of the regression models suggests that relevant factors of digital vulnerability may remain outside the adopted framework. In particular, the absence of variables related to digital platform architecture or interface design prevents the study from fully capturing the mechanisms through which digital financial behaviour is shaped.

5.3 Directions for Further Research or Improvements

Building upon the findings of this study, several relevant directions for future research emerge, particularly in relation to longitudinal data availability, methodological extensions, and the integration of additional explanatory dimensions. First, an important opportunity is represented by future waves of the IACOFI survey. Since the 2023 edition constitutes the first wave including indicators specifically related to digital financial literacy, future data collections may enable longitudinal comparisons over time. This would allow the identification of evolving trends in DFL profiles and the assessment of whether policy interventions aimed at reducing the detected structural weaknesses produce measurable improvements across the population. In this perspective, future research could move beyond static associations and investigate the persistence and evolution of digital financial literacy profiles.

Furthermore, a second relevant direction concerns the integration of additional explanatory dimensions to better capture the broader mechanisms underlying digital financial vulnerability. Future surveys may expand the measurement of digital financial literacy by including additional dimensions and a broader set of psychological and behavioural indicators. At the same time, the limited explanatory power of the behavioural mismatch regressions suggests the need to progressively shift the analytical focus beyond individual-level socio-demographic characteristics toward structural and environmental determinants of digital financial behaviour. In this perspective, future empirical frameworks could include variables related to digital platform architecture, user interface design, default security settings, and exposure to platform features that encourage or discourage secure digital behaviour.

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